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$$\Rightarrow x^2 = 4a^2 \cot^2\alpha \quad \ldots(1)$$

and

$$y = (v_0 \sin\alpha)t - \frac{1}{2}gt^2 = (v_0 \sin\alpha)\left(\frac{2a}{v_0 \sin\alpha}\right) - \frac{1}{2}g\left(\frac{2a}{v_0 \sin\alpha}\right)^2$$

$$= 2a - \frac{1}{2}g\frac{4a^2}{v_0^2 \sin^2\alpha} = 2a - g\frac{2a^2}{(2ga)\sin^2\alpha} = 2a - a\csc^2\alpha$$

$$= 2a - a(1+\cot^2\alpha) = 2a - a - a\cot^2\alpha = a - a\cot^2\alpha$$

$$\Rightarrow 4ay = 4a^2 - 4a^2\cot^2\alpha \quad \ldots(2)$$

Adding (1) and (2), we have

$$x^2 + 4ay = 4a^2\cot^2\alpha + 4a^2 - 4a^2\cot^2\alpha$$

$$x^2 = -4ay + 4a^2$$

$$x^2 = -4a(y-a) \quad \ldots(3)$$

This shows that at time $t = \frac{2a}{V}$ the particle is on a fixed parabola (parabola of safety).

Let θ be the direction of motion. Differentiating (3) w.r.t. x we have

$$2x = -4a\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{2a}$$

$$\Rightarrow \tan\theta = -\frac{2a\cot\alpha}{2a}$$

$$\Rightarrow \tan\theta = -\cot\alpha$$

$$\Rightarrow \tan\theta = \tan\left(\alpha + \frac{\pi}{2}\right)$$

$$\Rightarrow \theta = \alpha + \frac{\pi}{2}$$

This shows that the direction of motion of particle is then perpendicular to its direction of projection.

EXERCISE 9

Multiple Choice Questions (MCQs)

Four options are given in each of the following questions fill the circle in front of that choice which you think is correct. Cutting or filling two or more circles is not allowed:

Q.1

(i) If a particle is projected making a certain angle with horizontal into the air with a certain velocity or dropped from a moving plane then it is called a

(a) projectile (b) hyperbola (c) circle (d) straight line

ⓐ ⓑ ⓒ ⓓ

(ii) The path of a projectile is called its

(a) velocity (b) acceleration (c) trajectory (d) none of these

ⓐ ⓑ ⓒ ⓓ

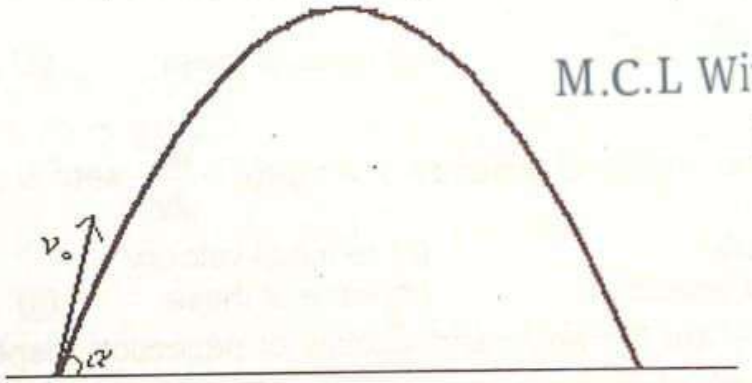
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(iii) Angle with which the particle is projected is called the _____ of projection.
(a) velocity (b) acceleration (c) angle (d) none of these
ⓐ ⓑ ⓒ ⓓ

(iv) The velocity with which the particle is projected is called the _____ of projection.
(a) velocity (b) acceleration (c) angle (d) none of these
ⓐ ⓑ ⓒ ⓓ

(v) If we neglect the air resistance, a projectile is a particle moving under the influence of _____ force.
(a) frictional (b) gravitational (c) angle (d) none of these
ⓐ ⓑ ⓒ ⓓ

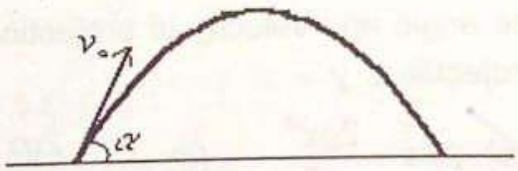
(vi) In the figure below, the horizontal component of velocity of projection is



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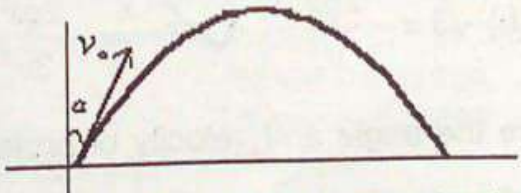
(a) $v_\circ \cos\alpha$ (b) $v_\circ \sin\alpha$ (c) $v_\circ \tan\alpha$ (d) $v_\circ \cot\alpha$
ⓐ ⓑ ⓒ ⓓ

(vii) In the figure below,



the vertical component of velocity of projection is
(a) $v_\circ \cos\alpha$ (b) $v_\circ \sin\alpha$ (c) $v_\circ \tan\alpha$ (d) $v_\circ \cot\alpha$
ⓐ ⓑ ⓒ ⓓ

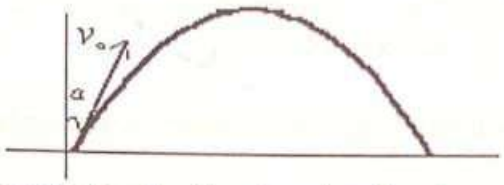
(viii) In the figure below,



the horizontal component of velocity of projection is
(a) $v_\circ \cos\alpha$ (b) $v_\circ \sin\alpha$ (c) $v_\circ \tan\alpha$ (d) $v_\circ \cot\alpha$
ⓐ ⓑ ⓒ ⓓ

(ix) In the figure below,

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the vertical component of velocity of projection is

(a) $v_\circ \cos\alpha$ (b) $v_\circ \sin\alpha$ (c) $v_\circ \tan\alpha$ (d) $v_\circ \cot\alpha$

ⓐ ⓑ ⓒ ⓓ

(x) The equation of path of projectile is $y =$

(a) $x\sin\alpha - \frac{gx^2}{2v_\circ^2}\sec^2\alpha$ (b) $x\tan\alpha - \frac{gx^2}{2v_\circ^2}\sec^2\alpha$

(c) $x\tan\alpha - \frac{gx^2}{2v_\circ^2}\cos^2\alpha$ (d) none of these ⓐ ⓑ ⓒ ⓓ

(xi) In the equation of path of projectile $y = x\tan\alpha - \frac{gx^2}{2v_\circ^2}\sec^2\alpha$, $v_\circ$ is the

(a) final velocity (b) terminal velocity
(c) velocity of projection (d) none of these ⓐ ⓑ ⓒ ⓓ

(xii) If 45° and $v_\circ$ are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$

(a) $x - \frac{gx^2}{v_\circ^2}$ (b) $\sqrt{3}\,x - \frac{2gx^2}{v_\circ^2}$ (c) $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_\circ^2}$ (d) $x - gv_\circ^2 x^2$

ⓐ ⓑ ⓒ ⓓ

(xiii) If 60° and $v_\circ$ are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$

(a) $x - \frac{gx^2}{v_\circ^2}$ (b) $\sqrt{3}\,x - \frac{2gx^2}{v_\circ^2}$ (c) $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_\circ^2}$ (d) $x - gv_\circ^2 x^2$

ⓐ ⓑ ⓒ ⓓ

(xiv) If 30° and $v_\circ$ are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$

(a) $x - \frac{gx^2}{v_\circ^2}$ (b) $\sqrt{3}\,x - \frac{2gx^2}{v_\circ^2}$ (c) $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_\circ^2}$ (d) $x - gv_\circ^2 x^2$

ⓐ ⓑ ⓒ ⓓ

(xv) If 45° and $\frac{1}{3}$ m/s are the angle and velocity of projection respectively, then the equation of path of projectile is $y =$

(a) $x - \frac{gx^2}{v_\circ^2}$ (b) $\sqrt{3}\,x - \frac{2gx^2}{v_\circ^2}$ (c) $\frac{x}{\sqrt{3}} - \frac{2gx^2}{3v_\circ^2}$ (d) $x - 9gx^2$

ⓐ ⓑ ⓒ ⓓ

(xvi) The path of a projectile is a

(a) straight line (b) circle (c) parabola (d) none of these

ⓐ ⓑ ⓒ ⓓ

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(xvii) The vertex of the path of projectile is

(a) $\left(\frac{v_\circ^2 \sin 2\alpha}{2g}, \frac{v_\circ^2 \sin^2 \alpha}{2g}\right)$ (b) $\left(\frac{v_\circ^2 \cos 2\alpha}{2g}, \frac{v_\circ^2 \cos^2 \alpha}{2g}\right)$

(c) $\left(\frac{v_\circ^2 \sec 2\alpha}{2g}, \frac{v_\circ^2 \sec^2 \alpha}{2g}\right)$ (d) (0,0) ⓐ ⓑ ⓒ ⓓ

(xviii) If 30° and $v_\circ$ are the angle and velocity of projection respectively, then the vertex of the path of projectile is

(a) $\left(\frac{2\sqrt{3}v_\circ^2}{g}, \frac{v_\circ^2}{g}\right)$ (b) $\left(\frac{\sqrt{3}v_\circ^2}{g}, \frac{v_\circ^2}{2g}\right)$ (c) $\left(\frac{\sqrt{3}v_\circ^2}{2g}, \frac{v_\circ^2}{4g}\right)$ (d) $\left(\frac{\sqrt{3}v_\circ^2}{4g}, \frac{v_\circ^2}{8g}\right)$

ⓐ ⓑ ⓒ ⓓ

(xix) If 45° and $v_\circ$ are the angle and velocity of projection respectively, then the vertex of the path of projectile is

(a) $\left(\frac{2\sqrt{3}v_\circ^2}{g}, \frac{v_\circ^2}{g}\right)$ (b) $\left(\frac{v_\circ^2}{2g}, \frac{v_\circ^2}{4g}\right)$ (c) $\left(\frac{\sqrt{3}v_\circ^2}{4g}, \frac{3v_\circ^2}{8g}\right)$ (d) $\left(\frac{\sqrt{3}v_\circ^2}{4g}, \frac{v_\circ^2}{8g}\right)$

ⓐ ⓑ ⓒ ⓓ

(xx) If 60° and $v_\circ$ are the angle and velocity of projection respectively, then the vertex of the path of projectile is

(a) $\left(\frac{2\sqrt{3}v_\circ^2}{g}, \frac{v_\circ^2}{g}\right)$ (b) $\left(\frac{v_\circ^2}{2g}, \frac{v_\circ^2}{4g}\right)$ (c) $\left(\frac{\sqrt{3}v_\circ^2}{4g}, \frac{3v_\circ^2}{8g}\right)$ (d) $\left(\frac{\sqrt{3}v_\circ^2}{4g}, \frac{v_\circ^2}{8g}\right)$

ⓐ ⓑ ⓒ ⓓ

Q.2

(i) The latus rectum of the path of projectile is

(a) $\frac{2v_\circ^2 \sin^2 \alpha}{g}$ (b) $\frac{2v_\circ^2 \cos^2 \alpha}{g}$ (c) $\frac{v_\circ^2}{2g}$ (d) $2v_\circ^2 \sin^2 \alpha$

ⓐ ⓑ ⓒ ⓓ

(ii) The directrix of the path of projectile is

(a) $2v_\circ^2 \tan^2 \alpha$ (b) $2v_\circ^2 \cos^2 \alpha$ (c) $2v_\circ^2 \sin^2 \alpha$ (d) $\frac{v_\circ^2}{2g}$

ⓐ ⓑ ⓒ ⓓ

(iii) The distance covered along horizontal direction by the projectile from point of projection to the point where the projectile strikes the ground is called the

(a) horizontal range (b) vertical range
(c) horizontal velocity (d) none of these ⓐ ⓑ ⓒ ⓓ

(iv) The horizontal range of a projectile is $R =$

(a) $\frac{v_\circ^2 \tan 2\alpha}{g}$ (b) $\frac{v_\circ^2 \sin 2\alpha}{g}$ (c) $\frac{v_\circ^2 \cos 2\alpha}{g}$ (d) $v_\circ^2 \sin 2\alpha$

ⓐ ⓑ ⓒ ⓓ

(v) If 15° and $v_\circ$ are the angle and velocity of projection, then the horizontal range of projectile is

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(a) $\frac{v_\circ^2}{2g}$ (b) $\frac{\sqrt{3}v_\circ^2}{2g}$ (c) $\frac{v_\circ^2}{g}$ (d) $v_\circ^2 g$

ⓐ ⓑ ⓒ ⓓ

(vi) If 30° and $v_\circ$ are the angle and velocity of projection, then the horizontal range of projectile is

(a) $\frac{v_\circ^2}{2g}$ (b) $\frac{\sqrt{3}v_\circ^2}{2g}$ (c) $\frac{v_\circ^2}{g}$ (d) $v_\circ^2 g$

ⓐ ⓑ ⓒ ⓓ

(vii) The horizontal range of a projectile is maximum if angle of projection is

(a) 15° (b) 30° (c) 45° (d) 60°

ⓐ ⓑ ⓒ ⓓ

(viii) The maximum horizontal range of a projectile is

(a) $\frac{v_\circ^2}{2g}$ (b) $\frac{\sqrt{3}v_\circ^2}{2g}$ (c) $\frac{v_\circ^2}{g}$ (d) $v_\circ^2 g$

ⓐ ⓑ ⓒ ⓓ

(ix) If 7 ft/s is the velocity of projection, then the maximum horizontal range is

(a) $\frac{49}{32}$ ft (b) $\frac{9}{8}$ ft (c) $\frac{32}{49}$ ft (d) none of these

ⓐ ⓑ ⓒ ⓓ

(x) If 8 ft/s is the velocity of projection, then the maximum horizontal range is

(a) 6 ft (b) 4 ft (c) 3 ft (d) 2 ft

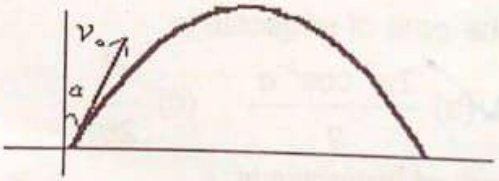
ⓐ ⓑ ⓒ ⓓ

(xi) If 80 ft/s is the velocity of projection, then the maximum horizontal range is

(a) 10 ft (b) 100 ft (c) 200 ft (d) 250 ft

ⓐ ⓑ ⓒ ⓓ

(xii) The horizontal range of the projectile shown in the figure below is



(a) $\frac{v_\circ^2 \sin 2\alpha}{g}$ (b) $\frac{v_\circ^2 \cos 2\alpha}{g}$ (c) $\frac{v_\circ^2 \tan 2\alpha}{g}$ (d) $v_\circ^2 g \sin 2\alpha\, t$

ⓐ ⓑ ⓒ ⓓ

(xiii) The maximum horizontal range of a projectile is 100 ft, the velocity of projection is

(a) 3.6 ft/s (b) 4.6 ft/s (c) 5.6 ft/s (d) 56.57 ft/s

ⓐ ⓑ ⓒ ⓓ

(xiv) The time of flight of a projectile is $t =$

(a) $\frac{v_\circ \sin\alpha}{2g}$ (b) $\frac{v_\circ \sin\alpha}{g}$ (c) $\frac{2v_\circ \sin\alpha}{g}$ (d) $2v_\circ g \sin\alpha$

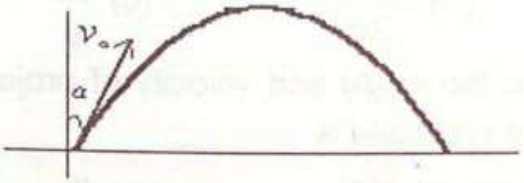
ⓐ ⓑ ⓒ ⓓ

(xv) The time taken to reach the maximum height of a projectile is

(a) $\frac{v_\circ \sin\alpha}{2g}$ (b) $\frac{v_\circ \sin\alpha}{g}$ (c) $\frac{2v_\circ \sin\alpha}{g}$ (d) $2v_\circ g \sin\alpha$

ⓐ ⓑ ⓒ ⓓ

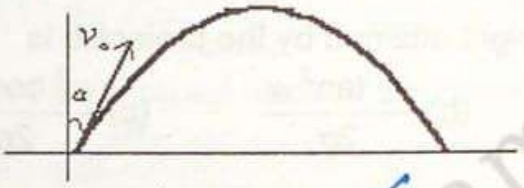
(xvi) The time of flight of the projectile shown in the figure below is



(a) $\frac{v_\circ \sin\alpha}{2g}$ (b) $\frac{v_\circ \sin\alpha}{g}$ (c) $\frac{2v_\circ \sin\alpha}{g}$ (d) $\frac{2v_\circ \cos\alpha}{g}$

ⓐ ⓑ ⓒ ⓓ

(xvii) The time taken by the projectile to attain the maximum height of the projectile in the figure below is



(a) $\frac{v_\circ \sin\alpha}{2g}$ (b) $\frac{v_\circ \sin\alpha}{g}$ (c) $\frac{v_\circ \cos\alpha}{g}$ (d) $\frac{2v_\circ \cos\alpha}{g}$

ⓐ ⓑ ⓒ ⓓ

(xviii) If 45° and $v_\circ$ are the angle and velocity of projection respectively, then the time of flight of the projectile is

(a) $\frac{\sqrt{2}v_\circ}{g}$ (b) $\frac{v_\circ}{g}$ (c) $\frac{\sqrt{3}v_\circ}{g}$ (d) $\frac{v_\circ}{\sqrt{2}g}$

ⓐ ⓑ ⓒ ⓓ

(xix) If 30° and $v_\circ$ are the angle and velocity of projection respectively, then the time of flight of the projectile is

(a) $\frac{\sqrt{2}v_\circ}{g}$ (b) $\frac{v_\circ}{g}$ (c) $\frac{\sqrt{3}v_\circ}{g}$ (d) $\frac{v_\circ}{\sqrt{2}g}$

ⓐ ⓑ ⓒ ⓓ

(xx) If 60° and $v_\circ$ are the angle and velocity of projection respectively, then the time of flight of the projectile is

(a) $\frac{\sqrt{2}v_\circ}{g}$ (b) $\frac{v_\circ}{g}$ (c) $\frac{\sqrt{3}v_\circ}{g}$ (d) $\frac{v_\circ}{\sqrt{2}g}$

ⓐ ⓑ ⓒ ⓓ

Q.3

(i) If 45° and $v_\circ$ are the angle and velocity of projection respectively, then the time taken to attain the maximum height of the projectile is

(a) $\frac{\sqrt{2}v_\circ}{g}$ (b) $\frac{v_\circ}{g}$ (c) $\frac{\sqrt{3}v_\circ}{g}$ (d) $\frac{v_\circ}{\sqrt{2}g}$

ⓐ ⓑ ⓒ ⓓ

(ii) If 30° and $v_\circ$ are the angle and velocity of projection respectively, then the time taken to attain the maximum height of the projectile is

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(a) $\frac{v_\circ}{2g}$ (b) $\frac{\sqrt{3}\,v_\circ}{2g}$ (c) $\frac{v_\circ}{g}$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

(iii) If 60° and $v_\circ$ are the angle and velocity of projection respectively, then the time of flight of the projectile is

(a) $\frac{v_\circ}{2g}$ (b) $\frac{\sqrt{3}\,v_\circ}{2g}$ (c) $\frac{v_\circ}{g}$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

(iv) A projectile is projected with speed 30 m/s at an angle of 30° to the horizontal, the time of flight is

(a) 4 s (b) 3 s (c) 2 s (d) 1 s

ⓐ ⓑ ⓒ ⓓ

(v) The maximum height attained by the projectile is

(a) $\frac{v_\circ^2 \sec^2\alpha}{2g}$ (b) $\frac{v_\circ^2 \tan^2\alpha}{2g}$ (c) $\frac{v_\circ^2 \cos^2\alpha}{2g}$ (d) $\frac{v_\circ^2 \sin^2\alpha}{2g}$

ⓐ ⓑ ⓒ ⓓ

(vi) The speed at a point P of the projectile is _____ as attained by a particle of the same mass at P falling freely from the point vertically above P on the directrix of the trajectory.

(a) less than that (b) same
(c) greater than that (d) none of these

ⓐ ⓑ ⓒ ⓓ

(vii) The least speed to hit the projectile a point (a,b) is

(a) $\sqrt{g(b+\sqrt{a^2+b^2})}$ (b) $g(b+\sqrt{a^2+b^2})$

(c) $[g(b+\sqrt{a^2+b^2})]^2$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

(viii) The least speed to hit the projectile a point $(1,2)$ is

(a) $g(2+\sqrt{5})$ (b) $\sqrt{g(2+\sqrt{5})}$ (c) $3\sqrt{g}$ (d) $3g$

ⓐ ⓑ ⓒ ⓓ

(ix) The least speed to hit the projectile a point $(3,4)$ is

(a) $g(2+\sqrt{5})$ (b) $\sqrt{g(2+\sqrt{5})}$ (c) $3\sqrt{g}$ (d) $3g$

ⓐ ⓑ ⓒ ⓓ

(x) The range on an inclined plane of inclination β is

(a) $\frac{2v_\circ^2 \cos\alpha \sin(\alpha-\beta)}{g\sin^2\beta}$ (b) $\frac{2v_\circ^2 \cos\alpha \sin(\alpha-\beta)}{g\cos^2\beta}$

(c) $\frac{2v_\circ^2 \cos\alpha \sin(\alpha-\beta)}{g\tan^2\beta}$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

(xi) The maximum range on an inclined plane of inclination β is

(a) $\frac{v_\circ^2}{g(1+\sin\beta)}$ (b) $\frac{v_\circ^2}{2g(1+\sin\beta)}$ (c) $\frac{v_\circ^2}{3g(1+\sin\beta)}$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

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(xii) The maximum range on an inclined plane of inclination $\beta = 30°$ is

(a) $\frac{v_\circ^2}{3g}$ (b) $\frac{2v_\circ^2}{3g}$ (c) $\frac{v_\circ^2}{g}$ (d) $gv_\circ^2$

ⓐ ⓑ ⓒ ⓓ

(xiii) The maximum range down the inclined plane of inclination $-\beta$ is

(a) $\frac{v_\circ^2}{g(1+\sin\beta)}$ (b) $\frac{v_\circ^2}{2g(1+\sin\beta)}$ (c) $\frac{v_\circ^2}{3g(1+\sin\beta)}$ (d) $\frac{v_\circ^2}{g(1-\sin\beta)}$

ⓐ ⓑ ⓒ ⓓ

(xiv) The time of flight on an inclined plane of inclination β is $t =$

(a) $\frac{2v_\circ\cos(\alpha+\beta)}{g\cos\beta}$ (b) $\frac{2v_\circ\sin(\alpha+\beta)}{g\cos\beta}$ (c) $\frac{2v_\circ\sin(\alpha-\beta)}{g\cos\beta}$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

(xv) Time of flight down the inclined plane of inclination $-\beta$ is $t =$

(a) $\frac{2v_\circ\cos(\alpha+\beta)}{g\cos\beta}$ (b) $\frac{2v_\circ\sin(\alpha+\beta)}{g\cos\beta}$ (c) $\frac{2v_\circ\sin(\alpha-\beta)}{g\cos\beta}$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

(xvi) The locus of all points on different trajectories which lie at maximum distances from point of projection is called _____ of safety.

(a) hyperbola (b) ellipse (c) parabola (d) circle

ⓐ ⓑ ⓒ ⓓ

(xvii) The equation of parabola of safety is $x^2 =$

(a) $-\frac{2v_\circ^2}{g}\left(y-\frac{2v_\circ^2}{g}\right)$ (b) $-\frac{2v_\circ^2}{g}\left(y-\frac{v_\circ^2}{g}\right)$ (c) $-\frac{2v_\circ^2}{g}\left(y-\frac{v_\circ^2}{2g}\right)$

(d) none of these

ⓐ ⓑ ⓒ ⓓ

(xviii) The directrix of the parabola of safety is

(a) $\frac{v_\circ^2}{4g}$ (b) $\frac{v_\circ^2}{2g}$ (c) $\frac{v_\circ^2}{g}$ (d) $\frac{2v_\circ^2}{g}$

ⓐ ⓑ ⓒ ⓓ

(xix) The vertex of the parabola of safety is

(a) $\left(0,\frac{4v_\circ^2}{g}\right)$ (b) $\left(0,\frac{2v_\circ^2}{g}\right)$ (c) $\left(0,\frac{v_\circ^2}{g}\right)$ (d) $\left(0,\frac{v_\circ^2}{2g}\right)$

ⓐ ⓑ ⓒ ⓓ

(xx) The focus of the parabola of safety is

(a) (1,2) (b) (2,1) (c) (1,1) (d) (0,0)

ⓐ ⓑ ⓒ ⓓ

Short Questions

Q.4 Solve / write answers of the following short questions:

(i) What is a projectile?

(ii) If 8 ft/s is the velocity of projection, then find the maximum horizontal range.

(iii) If 80 ft/s is the velocity of projection, then find the maximum horizontal range.

(iv) The maximum horizontal range of a projectile is 100 ft. Find the velocity of projection.

(v) A projectile is projected with speed 30 m/s at an angle of 30° to the horizontal, find the time of flight.

(vi) A projectile is projected with speed 30 m/s at an angle of 30° to the horizontal, find the maximum height of the projectile.

(vii) A projectile is projected with speed 40 m/s at an angle of 45° to the horizontal, find the time of flight.

(viii) A projectile is projected with speed 35 m/s at an angle of 60° to the horizontal, find the time of flight.

(ix) A ball is thrown with a speed of 20 m/s and hits the horizontal ground 20 m away. Find the angle of projection.

(x) A ball is thrown with a speed of 30 m/s and hits the horizontal ground 30 m away. Find the angle of projection.

(xi) Find the range of a projectile on an inclined plane of inclination 30° if 60° and $v_\circ$ are the angle and speed of projection respectively.

(xii) Find the maximum range on an inclined plane of inclination $\beta = 30°$.

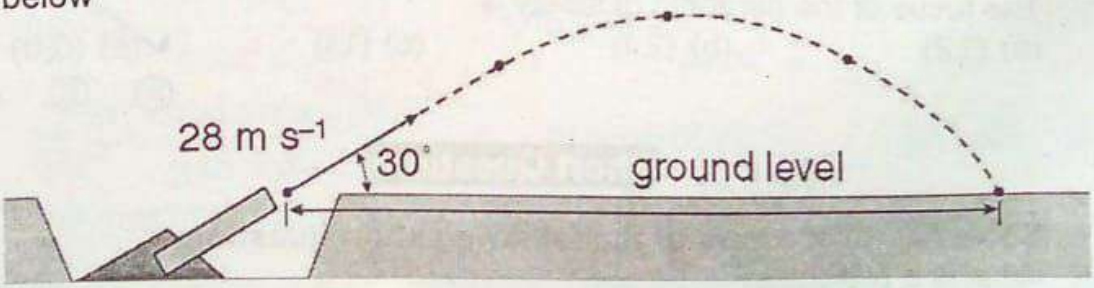
(xiii) If R_1 and R_2 are the ranges of a projectile when it is projected at angles 30° and 75° respectively with the same speed of projection. Show that $R_1 = \sqrt{3}R_2$.

(xiv) If R_1 and R_2 are the ranges of a projectile when it is projected at angles 15° and 75° respectively with the same speed of projection. Show that $R_1 = R_2$.

(xv) Find the equation of parabola of safety of a particle if it is projected with a speed 10 ft/s.

(xvi) Find the equation of parabola of safety of a particle if it is projected with a speed 5 ft/s.

(xvii) A senior physics class conducting a research project on projectile motion constructs a device that can launch a cricket ball. The launching device is designed so that the ball can be launched at ground level with an initial velocity of 28 m/s at an angle of 30° to the horizontal as shown in figure below



Calculate the horizontal component of the velocity of the ball:

(a) initially (b) after 1.0 s (c) after 2.0 s

SUMMARY

- A projectile is any object that once projected or dropped continues in motion by its own mass and is influenced only by the downward force of gravity.
- The maximum distance covered along horizontal direction by a projectile during its flight is called the horizontal range.
- The time taken by a projectile during its flight form point of projection to the point where it strikes the ground is called the time of flight.
- The speed at a point P of the projectile is the same as attained by a particle of the same mass at P falling freely from the point vertically above P on the directrix of the trajectory.
- At any point P on the parabolic trajectory of the projectile, the kinetic energy is the same as acquired by it in falling freely under gravity to P from that point on the directrix which is vertically above P.
- The least velocity with which a particle must be projected so that it passes through the points P and Q at the heights h_p and h_q respectively from the ground is given by $\sqrt{g(h_p + h_q + PQ)}$.
- The locus of all points on different trajectories which lie at maximum distances from point of projection is called parabola of safety.
- Equation of path of projectile is $y = x\tan\alpha - \dfrac{gx^2}{2v_0^2}\sec^2\alpha$.
- Maximum height attained by the projectile is $H_{\max} = \dfrac{v_0^2 \sin^2\alpha}{2g}$.
- Horizontal range of the projectile is $R = \dfrac{v_0^2 \sin 2\alpha}{g}$.
- Maximum horizontal range of the projectile is $R_{\max} = \dfrac{v_0^2}{g}$.
- Time of flight of the projectile is $t = \dfrac{2v_0 \sin\alpha}{g}$.
- Maximum range on an inclined plane is $R_{\max} = \dfrac{v_0^2}{g(1+\sin\beta)}$.
- Time of flight on an inclined plane is $t = \dfrac{2v_0 \sin(\alpha - \beta)}{g\cos\beta}$.
- Path of parabola of safety is $x^2 = -\dfrac{2v_0^2}{g}\left(y - \dfrac{v_0^2}{2g}\right)$.

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EXERCISE 10

Multiple Choice Questions (MCQs)

Four options are given in each of the following questions fill the circle in front of that choice which you think is correct. Cutting or filling two or more circles is not allowed:

Q.1

(i) A force acting on a particle is said to be _____ if its line of action always passes through a fixed point.
(a) applied force (b) friction (c) central force (d) none of these
ⓐ ⓑ ⓒ ⓓ

(ii) The central force is _____ if it is directed towards its centre.
(a) attractive (b) not attractive (c) repulsive (d) none of these
ⓐ ⓑ ⓒ ⓓ

(iii) The central force is _____ if it is directed away from the centre.
(a) attractive (b) repulsive (c) not repulsive (d) none of these
ⓐ ⓑ ⓒ ⓓ

(iv) Orbit of a particle moving under a central force is necessarily _____.
(a) a space curve (b) a plane curve (c) a straight line (d) none of these
ⓐ ⓑ ⓒ ⓓ

(v) The orbit described under a central attractive force varying directly as the distance is _____ having the centre at the centre of the force.
(a) an ellipse (b) a circle (c) a straight line (d) none of these
ⓐ ⓑ ⓒ ⓓ

(vi) A central force $\vec{F}$ is said to be conservative force if $\nabla \times \vec{F} =$
(a) 3 (b) 2 (c) 1 (d) 0
ⓐ ⓑ ⓒ ⓓ

(vii) In a conservative field, the total energy of a particle _____ throughout the motion.
(a) varies rapidly (b) does not remain constant
(c) remains constant (d) none of these ⓐ ⓑ ⓒ ⓓ

(viii) A central force is conservative if and only if it is a function of
(a) velocity and acceleration (b) distance and velocity
(c) distance alone (d) none of these ⓐ ⓑ ⓒ ⓓ

(ix) Areal speed of the radius vector $\vec{r}$ joining the particle to the centre of force is
(a) not constant (b) constant
(c) variable (d) none of these ⓐ ⓑ ⓒ ⓓ

(x) The component of force along the radius vector is

(a) $F_r = m\left[\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right]$ (b) $F_\theta = \frac{m}{r}\frac{d}{dt}\left(r^2\frac{d\theta}{dt}\right)$

(c) $F_r = m\left[\frac{dr}{dt} - r\left(\frac{d\theta}{dt}\right)\right]$ (d) none of these ⓐ ⓑ ⓒ ⓓ

(xi) The component of force perpendicular to the radius vector is

(a) $F_r = m\left[\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right]$ (b) $F_\theta = \frac{m}{r}\frac{d}{dt}\left(r^2\frac{d\theta}{dt}\right)$

(c) $F_r = m\left[\frac{dr}{dt} - r\left(\frac{d\theta}{dt}\right)\right]$ (d) none of these ⓐ ⓑ ⓒ ⓓ

(xii) When the force $\vec{F}$ is an attractive force directed towards origin, then $F_r = -F$ and $F_\theta =$

(a) 3 (b) 2 (c) 1 (d) 0 ⓐ ⓑ ⓒ ⓓ

(xiii) The differential equation of the orbit in polar coordinates is $\frac{d^2u}{d\theta^2} + u =$

(a) $\frac{f}{hu}$ (b) $\frac{f}{h^2u^2}$ (c) $\frac{f}{h^3u^3}$ (d) none of these ⓐ ⓑ ⓒ ⓓ

(xiv) The length of the radius vector at an apse is known as the apsidal

(a) acceleration (b) velocity (c) distance (d) none of these ⓐ ⓑ ⓒ ⓓ

(xv) The line joining an apse to the centre of force is _____ line.

(a) a focal line (b) an apse (c) an initial (d) none of these ⓐ ⓑ ⓒ ⓓ

(xvi) The angle between two consecutive apse lines is an

(a) angle of friction (b) angle of projection
(c) apsidal angle (d) angle of depression ⓐ ⓑ ⓒ ⓓ

(xvii) The analytic condition for a point with radius vector r to be an apse is $\frac{dr}{d\theta} =$

(a) 1 (b) 0 (c) −1 (d) −2 ⓐ ⓑ ⓒ ⓓ

(xviii) If ϕ is the angle between the radius vector and the tangent at some point, then $\tan\phi =$

(a) $r\frac{d\theta}{dr}$ (b) $\frac{1}{r}\frac{dr}{d\theta}$ (c) $\frac{dr}{d\theta}$ (d) none of these ⓐ ⓑ ⓒ ⓓ

(xix) If ϕ is the angle between the radius vector and the tangent at some point, then $\cot\phi =$

(a) $r\frac{d\theta}{dr}$ (b) $\frac{1}{r}\frac{dr}{d\theta}$ (c) $\frac{dr}{d\theta}$ (d) none of these ⓐ ⓑ ⓒ ⓓ

(xx) Kepler's _____ Law states that each planet describes an ellipse with the sun at one focus.

(a) Third (b) Second (c) First (d) none of these ⓐ ⓑ ⓒ ⓓ

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(xxi) Kepler's _____ Law states that the radius vector drawn from the sun to the planet sweeps out equal areas in equal times.
(a) Third (b) Second (c) First (d) none of these
ⓐ ⓑ ⓒ ⓓ

(xxii) Kepler's _____ Law states that the squares of the periods of the planets are proportional to the cubes of the semi-major axes of their orbits.
(a) Third (b) Second (c) First (d) none of these
ⓐ ⓑ ⓒ ⓓ

Short Questions

Q.2 Solve / write answers of the following short questions:

(i) State Kepler's First Law.

PU, 2013; 2011 (BS Math)

(ii) State Kepler's Second Law.

PU, 2013; 2011 (BS Math)

(iii) State Kepler's Third Law.

PU, 2013; 2011 (BS Math)

(iv) What is an apsidal angle?

(v) Define a central force.

(vi) When the central force is attractive; when it is repulsive?

(vii) What is an apse?

SUMMARY

- If the force $\vec{F}$ on a particle always passes through a fixed point O, the force is called a central force and O is called the centre of force.
- A central force is repulsive or attractive according as it is directed away from or towards the centre of force.
- The orbit of a particle moving under a central force is necessarily a plane curve.
- The orbit described under a central attractive force varying directly as the distance, is an ellipse having the centre at the centre of the force.
- When a particle moves under a central force, the areal velocity is constant.
- Differential equation of the orbit is $\frac{d^2u}{d\theta^2} + u = \frac{f}{h^2u^2}$.
- Differential equation of the orbit in pedal form is $\frac{h^2}{p^3}\frac{dp}{dr} = f$.
- An apse is a point on an orbit at which the radius vector drawn from the centre of force is a maximum or minimum.

- The length of the radius vector at such a point is known as the apsidal distance.
- The line joining an apse to the centre of force is an apse line.
- The angle between two consecutive apse lines is an apsidal angle.
- The radius vector through an apse and the tangent at it to the orbit are perpendicular.
- The planet will describe an ellipse, a parabola or a hyperbola according as total energy per unit mass, E, is negative, zero or positive.
- Kepler's First Law: Each planet describes an ellipse with the sun at one focus.
- Kepler's Second Law: The radius vector drawn from the sun to the planet sweeps out equal areas in equal times.
- Kepler's Third Law: The squares of the periods of the planets are proportional to the cubes of the semi-major axes of their orbits.

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Example-14: *Suppose that the motion of a falling spherical drop of liquid is opposed by a force proportional to the surface area of the drop to the density ρ of the medium through which it is falling and to nth power of its velocity. The density of the droplet is ρ_0 and it is assumed that the shape remains spherical. Show that the limiting velocity of the drop varies as the nth root of the radius of the drop. Gravity g is acting vertically downward.*

Solution: Let r and v be the radius and velocity of the drop respectively. Then $4\pi r^2$ is the surface area of the drop. Now force of resistance $\propto 4\pi r^2 \rho v^n$, so equation of motion is

$$m\ddot{x} = mg - 4k\pi r^2 \rho v^n, \qquad \text{where } k \text{ is proportinality constant}$$

$$\ddot{x} = g - \frac{4k\pi r^2 \rho v^n}{m} \qquad ...(1)$$

Since limiting velocity is attained at $\ddot{x} = 0$, so from (1), we have

$$g - \frac{4k\pi r^2 \rho v_{\lim}^n}{m} = 0$$

$$\Rightarrow 4k\pi r^2 \rho v_{\lim}^n = gm$$

$$\Rightarrow v_{\lim}^n = \frac{gm}{4k\pi r^2 \rho}$$

$$\Rightarrow v_{\lim} = \left(\frac{gm}{4k\pi r^2 \rho}\right)^{\frac{1}{n}} \qquad ...(2)$$

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

$$\rho_0 = \frac{m}{\frac{4}{3}\pi r^3} = \frac{3m}{4\pi r^3}$$

$$m = \frac{4}{3}\pi r^3 \rho_0$$

Putting this value in (2), we ehave

$$v_{\lim} = \left(\frac{4g\pi r^3 \rho_0}{3 \times 4k\pi r^2 \rho}\right)^{\frac{1}{n}} = \left(\frac{gr\rho_0}{3k\rho}\right)^{\frac{1}{n}}$$

$$v_{\lim} = \left(\frac{g\rho_0}{3k\rho}\right)^{\frac{1}{n}} r^{\frac{1}{n}} \propto r^{\frac{1}{n}}$$

EXERCISE 11

Multiple Choice Questions (MCQs)

Four options are given in each of the following questions fill the circle in front of that choice which you think is correct. Cutting or filling two or more circles is not allowed:

Q.1

(i) If a particle of mass m is released from rest at an initial height above the ground assuming the resistance of air proportional to the square of the velocity, then its equation of motion is $m\ddot{x} =$

(a) $mg - m\lambda v^2$ (b) $mg - m\lambda v$ (c) $mg - m\lambda v^3$ (d) none of these

Ⓐ Ⓑ Ⓒ Ⓓ

(ii) If a particle of mass m is released from rest at an initial height above the ground assuming the resistance of air proportional to the velocity, then its equation of motion is $m\ddot{x} =$

(a) $mg - m\lambda v^2$ (b) $mg - m\lambda v$ (c) $mg - m\lambda v^3$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

(iii) If a particle of mass m is released from rest at an initial height above the ground assuming the resistance of air proportional to the cube of the velocity, then its equation of motion is $m\ddot{x} =$

(a) $mg - m\lambda v^2$ (b) $mg - m\lambda v$ (c) $mg - m\lambda v^3$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

(iv) The maximum velocity of a particle moving in a resisting medium is called _____ velocity.

(a) initial (b) escape (c) terminal (d) final

ⓐ ⓑ ⓒ ⓓ

(v) For limiting velocity, the acceleration of the particle is

(a) negative (b) zero (c) positive (d) none of these

ⓐ ⓑ ⓒ ⓓ

(vi) If a particle of mass m is projected vertically upward with some initial speed assuming the resistance of air proportional to the square of the velocity, then its equation of motion is $m\ddot{x} =$

(a) $-mg - m\lambda v^2$ (b) $mg + m\lambda v^2$ (c) $mg + m\lambda v$ (d) none of these

ⓐ ⓑ ⓒ ⓓ

(vii) The small damped case for $\ddot{x} + k\dot{x} + \omega^2 x = 0$ is that

(a) $k^2 - 4\omega^2 < 0$ (b) $k^2 - 4\omega^2 = 0$

(c) $k^2 - 4\omega^2 > 0$ (d) none of these ⓐ ⓑ ⓒ ⓓ

(viii) The over damped case for $\ddot{x} + k\dot{x} + \omega^2 x = 0$ is that

(a) $k^2 - 4\omega^2 < 0$ (b) $k^2 - 4\omega^2 = 0$

(c) $k^2 - 4\omega^2 > 0$ (d) none of these ⓐ ⓑ ⓒ ⓓ

(ix) The critically damped case for $\ddot{x} + k\dot{x} + \omega^2 x = 0$ is that

(a) $k^2 - 4\omega^2 < 0$ (b) $k^2 - 4\omega^2 = 0$

(c) $k^2 - 4\omega^2 > 0$ (d) none of these ⓐ ⓑ ⓒ ⓓ

SUMMARY

- Equation of motion of harmonic oscillator is $\ddot{x} + k\dot{x} + \omega^2 x = 0$.
- Equation of motion of damped forced oscillator is $\ddot{x} + k\dot{x} + \omega^2 x = F(t)$.
- Terminal velocity or limiting velocity of a particle moving in a resisting medium is the maximum velocity it can attain.

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